Related Works

Color histograms have long been used to index images. They are very easy to compute and have proven useful. However, the problem with color histograms is that they do not include any spatial information of colors and thus are very liable to false positives. Besides, when there is a large appearance change in the image (such as adding borders), the color histograms will have difficulty finding the similarity between the image and the changed image.

Several methods have been proposed to modify the color histograms. Pass and Zabih [1,2] use the color coherence vector(CCV) which partitions each color into a coherent portion and an incoherent portion. CCV is easy to compute and has been shown to work better than the color histograms. But it is still very vulnerable to such appearance changes as adding borders or frames. Hsu et al. [3] selects a set of dominant colors and then partitions the image into rectangular regions so that each region mainly contains one color. The similarity of two images is then determined by the degree of overlap between regions of the same color. Smith and Chang [4] use a different algorithm to partition the image into regions. Employing histogram back-projection [5], they allow one region to have multiple different colors and the colors of each region are represented by a binary color set. The binary color sets and their location information combined is the feature vector for an image. This feature vector can facilitate the region queries. Huang et al. [6,7] tries to encode the spatial correlation of color-bin pairs by the "color correlogram". These approaches usually can give a better result than color histograms. Nevertheless, they (esp. the ones involving image partition) are much more difficult to compute and often require more storage space for the feature vectors than the color histograms.

Our approach tries to find an efficient way to incorporate the spatial information of the colors into the feature of images. We introduce the variances of the colors into the feature vector. The variances of the colors are overall indicators about how the colors are spatially distributed in the images. Combining the variance and the number of pixels of each color, we compose the color variance vector(CVV). The CVV is very easy to compute. In fact, it can be computed in a single pass over the image. CVV is especially effective and efficient in finding the similarity between the image and the image with a border or frame because of the special properties of the variance. It can identify the cropped versions of the image with the original one very easily. CVV can also serve as a general-purposed feature vector in the image retrieval system.

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